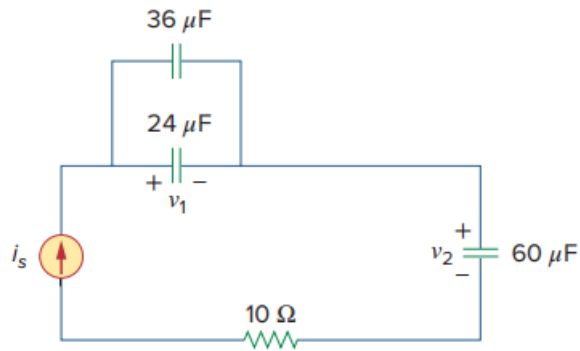


Problem

In the circuit in Fig. 6.64, let $i_s = 4.5e^{-2t}$ mA and the voltage across each capacitor is equal to zero at $t = 0$. Determine v_1 and v_2 and the energy stored in each capacitor for all $t > 0$.

Fig. 6.64:



Step-by-step solution

Step 1 of 8

(a)

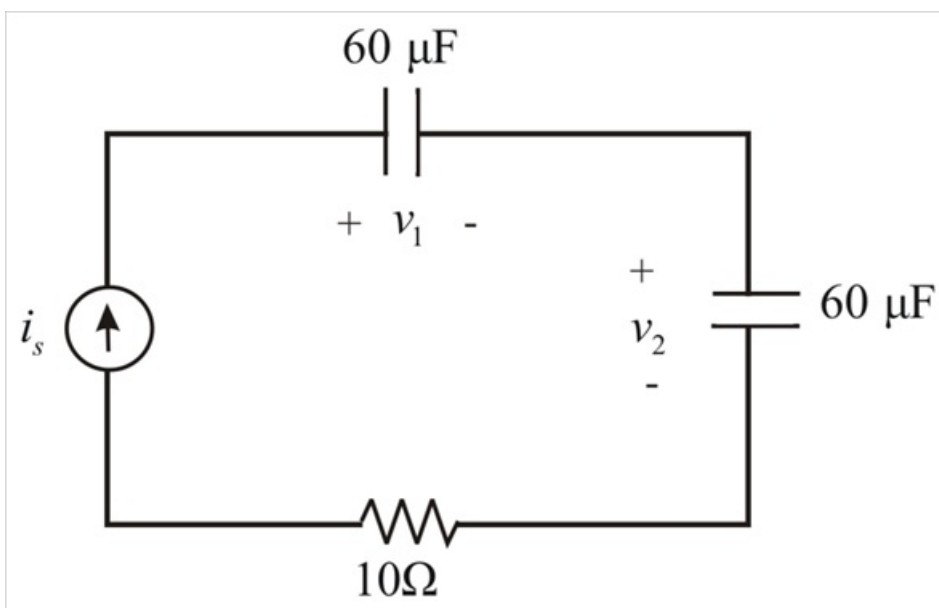
Refer to the Figure 6.64 in the textbook.

From the circuit $24\mu\text{F}$, $36\mu\text{F}$ capacitors are in parallel. So equivalent capacitance is,

$$\begin{aligned} C_{eq1} &= 36\mu\text{F} + 24\mu\text{F} \\ &= 60 \times 10^{-6} \\ &= 60 \times 10^{-6} \\ C_{eq1} &= 60\mu\text{F} \end{aligned}$$

Step 2 of 8

Draw the modified circuit.



Step 3 of 8

Refer to Figure 6.64 in the textbook.

Determine the value of $v_1(t)$.

Consider the general expression for the voltage across capacitor.

$$v_c(t) = \frac{1}{C} \int_0^t i_s(t) dt + v_c(0) \dots\dots (1)$$

Replace $60 \mu\text{F}$ for C , $4.5e^{-2t}$ mA for $i_s(t)$, 0 V for $v_c(0)$ in equation 1.

$$\begin{aligned} v_1(t) &= \frac{10^{-3}}{(60 \times 10^{-6})} \int_0^t 4.5e^{-2t} dt + 0 \\ &= \frac{4.5 \times 10^{-3}}{(60 \times 10^{-6})} \int_0^t e^{-2t} dt + 0 \\ &= 0.075 \times 10^3 \left(\frac{e^{-2t}}{-2} \right)_0^t + 0 \\ &= -0.0375 \times 10^3 (e^{-2t} - 1) \end{aligned}$$

Step 4 of 8

Further simplify the expression $v_1(t)$.

$$\begin{aligned} v_1(t) &= -0.0375 \times 10^3 (e^{-2t} - 1) + 0 \\ &= (-0.0375e^{-2t} + 0.0375) \times 10^3 + 0 \\ &= -37.5e^{-2t} + 37.5 \text{ V} \end{aligned}$$

Therefore, the voltage across $60 \mu\text{F}$ capacitor is $\boxed{-37.5e^{-2t} + 37.5 \text{ V}}$.

Step 5 of 8

Determine the value of $v_2(t)$.

Replace $60 \mu\text{F}$ for C , $4.5e^{-2t}$ mA for $i_s(t)$, 0 V for $v_c(0)$ in equation 1.

$$\begin{aligned} v_2(t) &= \frac{10^{-3}}{(60 \times 10^{-6})} \int_0^t 4.5e^{-2t} dt + 0 \\ &= \frac{4.5 \times 10^{-3}}{(60 \times 10^{-6})} \int_0^t e^{-2t} dt + 0 \\ &= 0.075 \times 10^3 \left(\frac{e^{-2t}}{-2} \right)_0^t + 0 \\ &= -0.0375 \times 10^3 (e^{-2t} - 1) \end{aligned}$$

Further simplify the expression $v_2(t)$.

$$\begin{aligned} v_2(t) &= -37.5(e^{-2t} - 1) + 0 \\ &= -37.5e^{-2t} + 37.5 + 0 \\ &= -37.5e^{-2t} + 37.5 \text{ V} \end{aligned}$$

Therefore, the voltage across $60 \mu\text{F}$ capacitor is $\boxed{-37.5e^{-2t} + 37.5 \text{ V}}$.

Step 6 of 8

(b)

Determine the energy in $60\text{ }\mu\text{F}$ capacitor at $t > 0$.

Consider the equation for energy stored in a capacitor.

$$w = \frac{1}{2} C (v(t))^2 \dots\dots (2)$$

$$\begin{aligned} w_{60\text{ }\mu\text{F}} &= \frac{1}{2} C v_1^2 \\ &= \frac{1}{2} \times 60 \times 10^{-6} \times v_1^2 \\ &= \frac{1}{2} \times 60 \times 10^{-6} \times \left(-0.0375 \times 10^3 (e^{-2t} - 1) \right)^2 \\ E_{60\text{ }\mu\text{F}} &= 0.042 (e^{-2t} - 1)^2 \text{ J} \end{aligned}$$

Step 7 of 8

Replace $36\text{ }\mu\text{F}$ for C , $-0.0375 \times 10^3 (e^{-2t} - 1)$ for $v(t)$ in equation 2.

$$\begin{aligned} w_{60} &= \frac{1}{2} (36 \times 10^{-6}) \left(-0.0375 \times 10^3 (e^{-2t} - 1) \right)^2 \\ E_{60\text{ }\mu\text{F}} &= 0.025 (e^{-2t} - 1)^2 \text{ J} \end{aligned}$$

Therefore, the energy stored in $36\text{ }\mu\text{F}$ capacitor is $\boxed{0.025 (e^{-2t} - 1)^2 \text{ J}}$.

Step 8 of 8

Replace $24\text{ }\mu\text{F}$ for C , $-0.0375 \times 10^3 (e^{-2t} - 1)$ for $v(t)$ in equation 2.

$$\begin{aligned} w_{60} &= \frac{1}{2} (24 \times 10^{-6}) \left(-0.0375 \times 10^3 (e^{-2t} - 1) \right)^2 \\ E_{60\text{ }\mu\text{F}} &= 0.017 (e^{-2t} - 1)^2 \text{ J} \end{aligned}$$

Therefore, the energy stored in $24\text{ }\mu\text{F}$ capacitor is $\boxed{0.017 (e^{-2t} - 1)^2 \text{ J}}$.